TEXAS A&M UNIVERSITY
Engineering

Kinetic Energy Recovery Systems

and

Life Cycle Management

of

Elevators

Report to KLEEMANN, Thessaloniki, Greece

For Completion of the Project Requirements for

Texas A&M University - ENGR410 Global Engineering Design Course

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Abstract

The world has committed to reducing greenhouse gas (GHG) emissions and limiting global warming. To achieve this goal, we must all work together. Various industries, including elevator manufacturers, are crucial in this decarbonization agenda. KLEEMANN is a well-known elevator manufacturer that provides products and services to several sectors worldwide, such as residential, office, shopping centers, hospitals, hotels, and cruise ships. All stakeholders associated with these industries share the responsibility of reducing GHG emissions. KLEEMANN is committed to decarbonizing elevator operations, lowering energy costs, improving lifecycle, and collaborating with stakeholders on the global decarbonization agenda, which promises significant benefits for GHG reduction.

Keywords: kinetic energy recovery systems, life cycle management, sustainable development goals, design thinking, flywheel, magnet coil system, smart manufacturing, zero defect manufacturing.

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Introduction and Background

Stavros Petridis, the Deputy Research and Development (R&D) Director of Electronics and Senior Manager; Dr. Ioanna Sfampa - Intellectual Property Specialist; and Mr. Ilias Matzarlis - Electronics R&D Software and Hardware Supervisor for KLEEMANN, headquartered in Thessaloniki Greece, are looking for support in engineering design. They want to implement kinetic energy recovery systems in vertical elevator systems and improve the lifecycle of elevators, which will reduce electrical power consumption and extend the life cycle based on predictive maintenance analysis.

Simos Simeonidis, the Manager of Simeonidis Tours, arranged a meeting on May 14th with Texas A&M University students, Professor Bjorn Bangstein, and KLEEMANN's Dr. Ioanna Sfampa and Mr. Ilias Matzarlis. The meeting focused on defining the scope and limitations of the student's participation in these projects' opportunities. This meeting was the first step in a timeline that included a visit to the KLEEMANN manufacturing facility, the design and analysis phase, and the final presentation of the projects.

The students visited the KLEEMANN manufacturing facility to gather information and familiarize themselves with the products, engineering design, and services. The visit started with the students having access to learn more about the products they will be working with. Dr. Ioanna Sfampa and Mr. Ilias Matzarlis introduced the projects. They encouraged students to ask questions to clarify the project expectations.

Sequentially, the students followed the design thinking and project management process to generate project charters and created an outline of the overall and individual projects. The group includes fourteen students from Texas A&M University and three from Aristotle University of Thessaloniki. The group was divided into two teams with diverse

engineering-focus backgrounds, with one local student being assigned to each team. This approach was considered the most effective way to address project activities and cultural learning opportunities.

Since the meeting, the students have been actively working on multiple designs and analyses. For instance, one team focuses on the energy efficiency of the KERS system. At the same time, another explores the potential impact on the elevator's lifecycle. These efforts meet the projects' expectations and enhance the students' global cultural understanding of engineering projects.

Purpose of the Projects

The purpose of these projects was to generate innovative ideas for kinetic energy recovery systems and the life cycle management of elevators produced by KLEEMANN. Both projects aimed to reduce overall energy consumption and lower the environmental impact of manufacturing, using, and recycling elevators. As a result, these projects support United Nations Sustainable Development Goals (SDG) 8, 9, 11, 12, and 13. See Figure 1.

Figure 1

United Nations Sustainable Development Goals (United Nations, 2023).

Significance of the Projects

These projects are significant because they support KLEEMANN's corporate responsibility and commitment to sustainability. KLEEMANN is dedicated to respecting its employees, customers, suppliers, business partners, and the end-users of its products. Hence, KLEEMANN emphasizes sustainable development and transparency in its operations. Therefore, their goal is to contribute positively to society and the environment, a commitment that is apparent in the students' project description.

Projects Approach

The overall project was divided into two individual projects: kinetic energy recovery systems and life cycle management. These projects aim to support sustainable development. Each project team was tasked with conducting independent research, design, and analysis. KLEEMANN is the project sponsor and provided overall direction and expectations. KLEEMANN also encouraged students to bring innovative ideas beyond their corporate limitations, such as time or budget constraints for implementation. Despite being separate projects, all students are committed to the overall goal and the individual projects; hence, these projects are presented together in one project report.

Engineering Design Process

The engineering design process is a systematic method used in various fields, such as product manufacturing, design, and research and development. Its goal is to create solutions for problems that satisfy provided constraints, resources, time, and budget. The engineering design process includes defined steps, usually involving problem identification, research, ideation, analysis of ideas, building, testing and refinement, and communication and reflection. In this project, the design thinking methodology, which emphasizes the user in the initial step, was incorporated into the engineering design and project management processes defined by the Project Management Institute.

Design Thinking Methodology

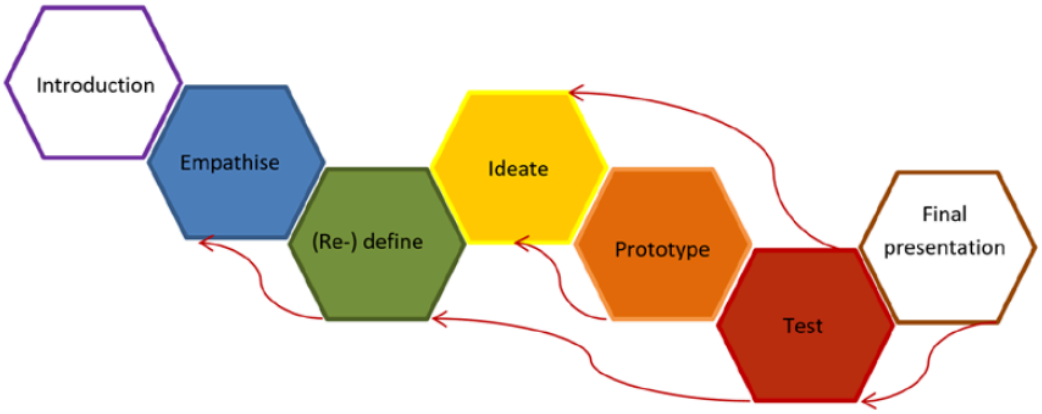
Design thinking is a methodology that provides a solution-based approach to solving problems. Design thinking is beneficial when used to tackle complex problems that are ill-defined or unknown because it serves to understand the human needs involved, reframe the problem in human-centric ways, create numerous ideas in brainstorming sessions, and adopt a hands-on approach to prototyping and testing (Pande & Bharathi, 2020). The design

thinking process consists of five core stages: Empathize, Define, Ideate, Prototype, and Test, see Figure 2.

The first stage, empathizing, involves teams conducting user-centric research to gain an empathetic understanding of the problem. Next, teams organize the gathered information and analyze observations to define the core problems in a human-centered design. In the ideate stage, teams generate ideas and use the gathered information to develop innovative solutions for the identified problems. Various ideation techniques can be employed. In the prototype stage, teams create prototypes based on the best ideas from the ideation phase. These prototypes are designed to test assumptions about stakeholders' needs. Finally, teams test the prototypes with real users to gather feedback and evaluate how well the solutions work. The test stage provides valuable insights for further refinement.

Figure 2

Design Thinking Model



Note. Adjusted for Texas A&M University ENGR410 Course. *(Grodzka, 2021)*

Project Management Process

Project management uses specific knowledge, skills, tools, and techniques to deliver value to stakeholders. All projects are a temporary effort to create value through a unique product, service, or result (PMBOK® Guide, 2021). All projects have a beginning, an end, a team, a budget, a schedule, and a set of expectations the team needs to meet. Each project is unique and differs from routine operations—the ongoing activities of an organization—because projects conclude once the goal is achieved (Pokharel, 2021). According to the Project Management Institute, a project management life cycle has five distinct phases. The five phases include conception and initiation, project planning, execution, performance/monitoring, and project closing (PMBOK® Guide, 2021). Each step is managed by individual processes with inputs and outputs. Each team in this project followed the steps laid out by PMI to follow industrially accepted project management for each respective project to be integrated into the final project deliverable to KLEEMANN.

Figure 3

Project Management Life Cycle

Note. (Bhandari, et al., 2022)

Project 1 - Kinetic Energy Recovery Systems

The primary goal of this project was to design a way to implement kinetic energy recovery systems (KERS) into elevator systems. Disaggregation and prioritization were used to form a logic tree (Figure 4) to define the following problem statement: KLEEMANN, an elevator manufacturer, is seeking a strategy to integrate KERS technology into their traction elevators to promote energy efficiency and cost-effectiveness, enhancing vertical transportation. In order to improve energy efficiency, two solutions were developed: harnessing rotational energy from the flywheel in the governor and creating a system of relative motion between a coil and magnet to produce electrical energy.

Figure 4

KERS Logic Tree

Background

KLEEMANN is seeking a solution to recover the kinetic energy lost in the process of elevator braking. The governing equation of kinetic energy losses, $\Delta\varsigma_T$, is

$$\Delta\varsigma_T = \frac{1}{2}m({v_f}^2 - {v_i}^2)$$

where m is the mass of the vehicle, v_f is the final velocity, and v_i is the initial velocity.

The governing equation for converting these kinetic energy losses to electrical energy is defined as

$$\frac{1}{2}m(v_f^{\ 2} - v_i^{\ 2}) = \frac{IVt}{\eta_g}$$

where m is the mass of the vehicle, v_f is the final velocity, v_i is the initial velocity, I is the current of the electric engine (as the role of a generator), V is the output voltage, t is time, and η_g is the efficiency of the energy conversion process (Armenta-Déu, C. 2023).

The defining equation for rotational kinetic energy, E_K, is

$$E_K = \frac{1}{2}I\omega^2$$

where I is moment of inertia for the given shape and ω is its angular velocity.

These three equations form the basis for a kinetic energy recovery system as it is applied to elevators. These equations are applied to the flywheel where the rotational energy is converted to electrical energy through a connection with a generator.

Flywheel Approach

For the flywheel design, the most optimal solution was found to be a solid disc made of cast iron material at the top of the elevator. The flywheel stores rotational energy as the elevator brakes and releases energy to kickstart the elevator's next movement. However, the flywheel can be connected to a generator to turn the rotational energy into electrical energy. The generator can be used to charge a battery for other uses, such as an extra power source for the building. Despite having lower energy storage capacity than a rimmed disc, the solid disc's advantages include higher strength and durability while maintaining lower maintenance requirements. To counter the lower storage of energy, the energy created can be recycled back into the system maintaining a cycle of energy throughout the elevator or the building.

Magnet and Coil System (MACS) Approach

Alternatively, there are several ideas for energy production using MACS, which are compared in Table 1. Technical diagrams for designs 2-3 are shown below the table, in Figures 6-7.

Potential Implementations:

1. A magnet moves through a separate, coil-wrapped tube in coordination with the movement of the elevator.
2. The counterweight's shaft is wrapped in conductive coil and magnets are attached to the counterweight.
3. A coil is extended from the sheave which rotates within a magnetic field.

Table 1

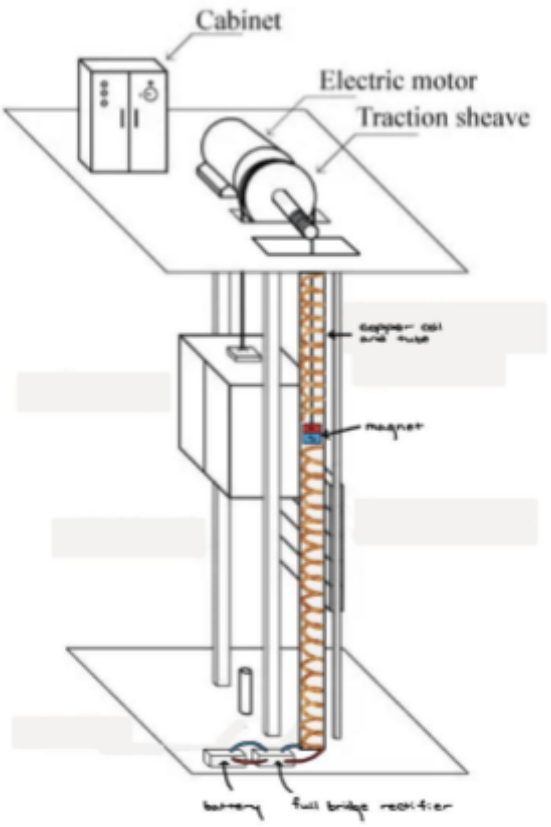
Comparison of different implementations of coil-magnet energy production system

Potential Implementation	Advantages	Disadvantages
1	● Cost-effective (compared to #2) ● A separate system makes it easier to add on to existing elevator systems ● Easier to maintain	● Requires a lot of wire (compared to #3) ● Need additional space ● Extra power may be required to move the weight of the magnet
2	● Magnet weight can be integrated into counterweight (would not need additional power)	● Requires a lot of wire (compared to #1 and #3) ● Since it is integrated into the elevator, would not be implemented as an "add-on"
3	● Less wire required (compared to #1 and #2) ● Cost-effective (compared to #2) ● No need to create a separate pulley system	● Must create spinning shaft off of motor (would not be implemented as an "add-on") ● Extra power is required to spin the coil

For the coil and magnet design, a neodymium magnet would move in conjunction with the elevator through an exterior tube wrapped in conductive coil to induce a current in the coil. This could be done by extending a bar from the sheave and attaching the magnet's cable to it. Therefore, as the elevator-counterweight system moves, the sheave and the bar would rotate, causing the exterior magnet to move vertically with it. This induced alternating current would run through a full bridge rectifier and capacitor to be converted to a DC voltage which would charge a battery stored in the elevator system as seen in Figure 5. The main advantage of the neodymium magnet is the high magnetic force per unit volume. An alternative option would be a ceramic magnet, which has a lower cost but is not strong enough to produce an induced voltage of significance for an elevator. Copper wire is a very common and inexpensive selection for most coils. However, its high conductivity and low resistivity can generate a very high current that can damage the equipment. Therefore, 80/20 nickel chrome alloy wire is a good choice in this case due to its higher resistivity and durability at a reasonable cost. For cost-effectiveness and easier maintenance access, the magnet would be attached to a separate cable connected to the same pulley system as the counterweight. The cable would move the magnet vertically through the conductive wire tube. The above diagram and description depict one formulation of this coil-magnet energy generation system.

Figure 5

Technical Diagram of Coil/Magnet System



In Figure 6, the same principles demonstrated in Figure 5 are applied; however, instead of using a separate system, the coils are directly wrapped around the counterweight. In this configuration, a magnet is attached to the counterweight itself. This design integrates the magnet and coil system more closely with the elevator's existing components, potentially reducing the need for additional space and simplifying the overall setup.

Figure 7 displays a slightly different approach, which utilizes the relative rotational motion between a stationary magnetic field and a spinning coil. As this coil would be axially mounted on the sheave, it would move as the elevator moves and its rotation within the magnetic field generates electrical energy. This energy can be collected and put back into the elevator system or into the building.

Figure 6

Technical Diagram of Coil/Magnet System (Design 2)

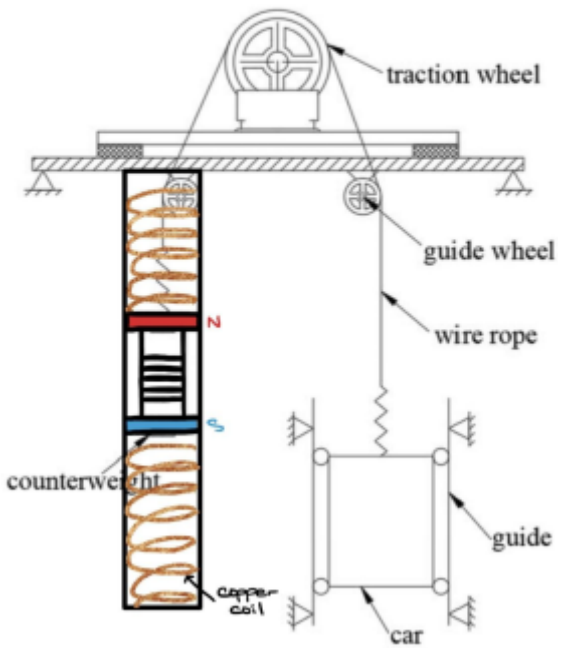
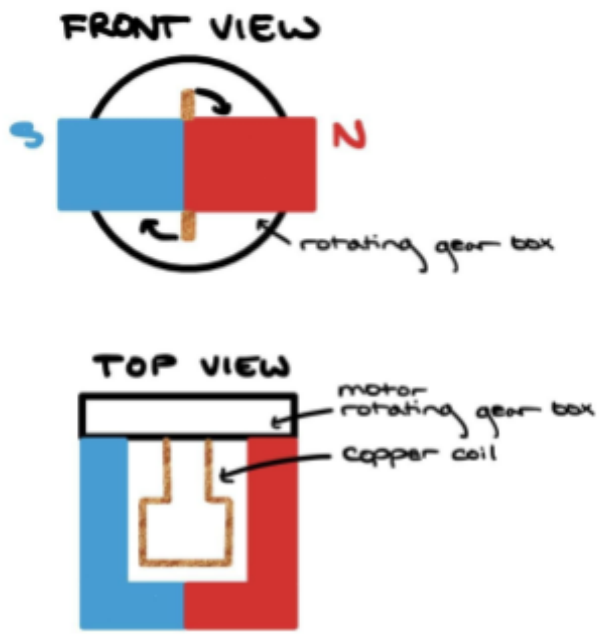


Figure 7

Technical Diagram of Coil/Magnet System (Design 3)



Rapid Design and Analysis of MACS

The design and analysis of the magnet and coil system (MACS) necessitated the generation of a testing framework that enabled the characterization of various configurations. As shown in Table 2, parameters such as the material, wire thickness, and radius were adjusted to determine potential improvements in the power output and cost constraints. By automating trade space exploration, an optimal solution could be rapidly identified. The program was implemented with object-oriented Python programming to streamline the analysis and promote modularity and scalability. The magnet and coil classes allow for the creation of objects with customized attributes to simulate the performance of various systems.

Table 2

Key Parameters for the MACS Analysis

Inputs
Magnet
Material
Magnetic Field Strength (T)
Magnet Radius (m)
Cost ($)
Coil
Material
Wire Thickness (m)
Coil Radius (m)
Number of Turns Per Meter
Cost Per Meter ($)
Resistivity (Ω·m)

Outputs
Power (W)
Voltage (V)
Current (A)
Resistance (Ω)
Wire Length (m)
Cost ($)

In order to determine the power output of the system, it was necessary to determine the induced electric potential. The electromotive force (EMF) generated in a coil due to a moving magnet is given by Faraday's Law of Induction:

$$\varepsilon =- N\frac{d\phi}{dt}$$

The magnetic flux, Φ, through a coil of cross-sectional area, A, in a uniform magnetic field, B, is:

$$\phi = BAcos(\theta)$$

Since the magnet moves linearly through the coil, the rate of change of magnetic flux can be expressed as:

$$\frac{d\phi}{dt} = BA$$

The induced EMF is therefore:

$$\varepsilon =- NBA$$

Using the EMF the current and power being produced can be found by:

$$I = \frac{\varepsilon}{R} \text{ and } P = \varepsilon I$$

where R is the resistance of the coil.

Recommendation

Based on the analysis conducted, one such implementation could be a commercial-off-the-shelf (COTS) 6” diameter neodymium magnet with a 3 meter long 0.08” diameter 80/20 Nickel Chrome alloy coil wire with 3000 turns. Assuming that one coil is integrated with an elevator moving at 1 meter per second, approximately 1350 W can be output. With the specifications provided by KLEEMANN, the five story building could have up to five coils to charge separate capacitors. The cost of the magnet and a single coil, excluding installation costs and electrical storage units, is nearly $1,700.

Project 2 - Life Cycle Management

Managing the lifecycle of products presents a significant challenge and opportunity for the manufacturing industry. Products must meet initial customer requirements and maintain efficiency throughout their operational life. Moreover, life cycle management (LCM) covers the entire product life cycle, from design initiation to operational life, disposal, and potential reuse (Westkämper, 2000). Global market trends, such as the sustainable use of natural resources, climate change, and new regulations, highlight the importance of LCM for manufacturers (Kamalakkannan et al., 2020).

The European Commission welcomed the provisional agreement reached on December 5, 2023, between the European Parliament and the Council on the Ecodesign for sustainable products regulation (Trincado et al., 2021). This forward-looking agreement aims to make sustainable products the new norm in the EU by ensuring that products last longer, use resources more efficiently, are more accessible for repair and recycling, contain fewer harmful substances, and include more recycled content, thereby paving the way for more sustainable use of energy and materials (Trincado et al., 2021).

The new agreement (law) builds on the Ecodesign Directive, which has successfully driven improved energy efficiency in products in the EU for almost 20 years (Kamalakkannan et al., 2020). It will allow the EU to progressively set performance and information requirements for essential products on the market.

The new Eco Design aims to boost circularity, covering topics:

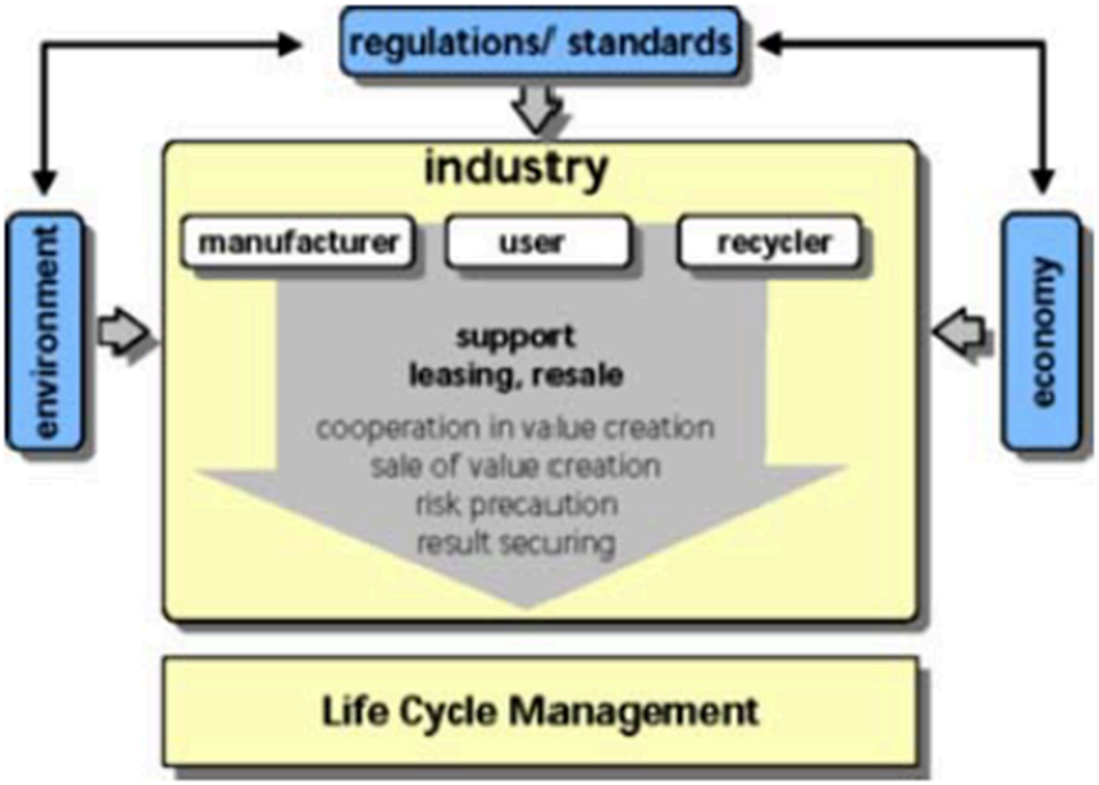
- product durability, reusability, upgradability, and repairability
- presence of chemical substances that inhibit the reuse and recycling of materials.
- energy and resource efficiency
- recycled content

- carbon and environmental footprints
- available product information, particularly a Digital Product Passport.

Based on the new provisional agreement, this project has focused on how LCM can help KLEEMANN comply with the new EU agreement for more sustainable, repairable, and circular products. LCM involves applicable regulations, product life cycles, and environmental impact assessments (see Figure 8).

Figure 8

Environment, Regulatory and Economy Demand for Life Cycle Management (Westkämper, 2000).



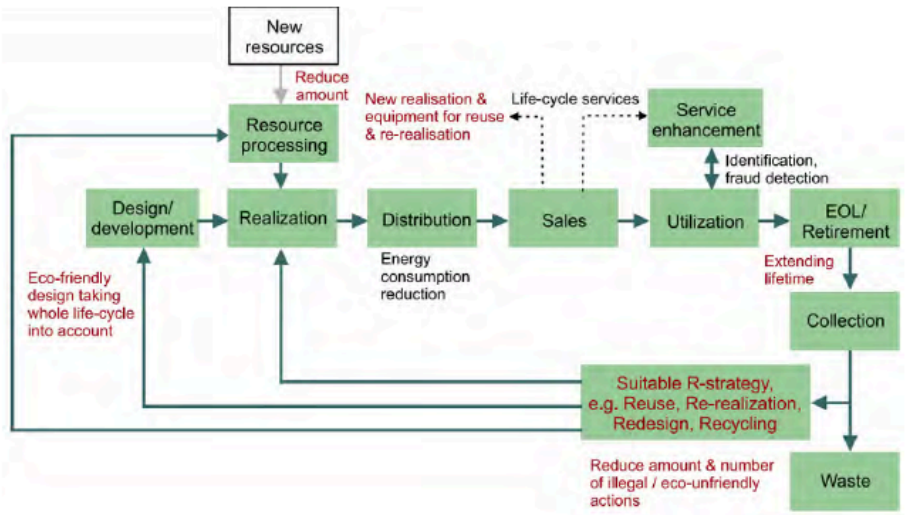
This "cradle-to-grave" approach covers sourcing materials, processing, transportation, manufacturing, distribution, use, reuse, maintenance, recycling, and disposal. LCM aims to

optimize a product's design, implementation, and life cycle activities in terms of costs and environmental impact (Westkämper, 2000). Continued advancement of their LCM can help KLEEMANN comply with the new provisional agreement. Furthermore, the advanced LCM practices can assist decision-makers at KLEEMANN in scaling up their product for the EU market and growing their business beyond the EU by promoting the benefits of LCM that include extended operational use, lower environmental impact, residual value of the elevators by refurbishing parts and sustainable disposal following completed use.

Moreover, the LCM can integrate circular economy aspects as defined by the Ellen MacArthur Foundation "to establish a restorative or regenerative industrial system based on sustainable development principles. It focuses on durability, maintenance, refurbishment, reuse, remanufacturing, and recycling of products to reduce the use of new materials and increase material productivity" (MacArthur, 2013) see Figure 9.

Figure 9

Life cycle model that considers the circular economy (Valkokari et al., 2016).



The circular economy approach is in line with LCM goals and has the potential to reduce global climate emissions significantly. Circular economy models include biological

and technical cycles, emphasizing the consumption and reuse of materials and products. Moreover, improved quality in manufacturing processes can lower environmental impacts. Hence, increasing a product's lifespan and zero-defect manufacturing (ZDM) are the proposed options from a resource efficiency standpoint. Extending a product's life cycle involves service solutions such as maintenance, repair, reuse, and remanufacturing, as well as design solutions that consider the requirements of a circular economy (Peng et al., 2022). From a circular economy perspective, the manufacturer must make eco-friendly decisions and include the whole life cycle of their elevators, starting at the design initiation to the end of the product life cycle, including recycling.

Initial Ideas

Throughout the ideation process, several different routes were taken to understand the best path to approach the problem. Addressing the sustainability aspect of life cycle management there were initial thoughts about implementing a sleep mode. The non-essential components, such as the cabin lights, fans, and display panels, are powered down when in sleep mode. To ensure no delays for the users, the elevator immediately wakes up when a button is pressed or a door is opened. Implementing this strategy would significantly decrease energy consumption during idle time, which typically accounts for up to 70% of the elevator’s lifetime and also limit the unnecessary wear and tear of the components. Aiming to further improve the products’ sustainability, another suggestion could be remanufacturing. Research indicates that remanufacturing is often viewed as a more environmentally preferable option compared to recycling. This preference is due to significant CO2 reductions, which account not only for the material production but also the manufacturing process itself. When an elevator is collected after being replaced or when a building is scheduled for demolition, the newer modules can be separated. These modules can then be

remanufactured to their original condition and recertified, thus extending their lifecycle and further reducing environmental impact.

Remanufacturing

The most important factor to keep in mind are the standards at which these components need to be upheld. Just to name a few examples that KLEEMANN already follows, the ISO 9001:2015 Quality Standard is essentially a generic standard requiring companies to consistently meet customer needs as well as regulatory requirements. There are also EU funding programs such as Horizon Europe, which is the primary EU funding program for research into more eco-friendly solutions for elevators, and there's the European Lift Directive 2014/33/EU, Annex XI (Module H1), which is a standard that increases the safety of the elevator.

While conducting research, remanufacturing seemed a promising option for elevator manufacturing as it is eco-friendly as well as cost-effective. Remanufacturing is the process of repairing parts rather than completely replacing them. Reusable materials could be collected and the end of an elevator's life and remanufactured to match the original condition. The key is to design products that can be easily disassembled into materials that can be reused, replaced or remanufactured. The elevators should be designed in a way that can be easily disassembled into parts that can be reused, replaced, and remanufactured.

If used parts remain compliant with the current safety standards and the integrity of the part remains, the part is able to be reused. Then, a process known as "reverse logistics" may be implemented. Reverse logistics is the process in the supply chain that deals with the return of any supplied goods to inventory. Establishing reverse logistics leads to an efficient collection of used products, which is critical for remanufacturing. Companies can take those newly gained raw materials from the product and put them back into production.

Various studies prove the viability of remanufacturing. One article wrote that remanufactured parts are 40% less expensive and reduce the carbon footprint by 30%. Another study led to the conclusion that retaining components such as the elevator counterweight, guide rails and landing doors when upgrading to a new, lighter, and smarter elevator saves up to 40% of embodied carbon emissions compared to a full replacement.

In conclusion, remanufacturing is cost effective, energy efficient, and environmentally friendly. Given the safety-critical nature of some elevator components, KLEEMANN could explore partnerships with specialized remanufacturing companies who have the expertise to refurbish components to as-new condition and certify them for reuse. Through research, the most effective way to reuse components is through converting them into raw materials to remanufacture in the future.

It's Important to engage with customers and help them understand the benefits of remanufacturing. Remanufactured components can offer cost savings and environmental benefits for customers. KLEEMANN could actively communicate these advantages and work with major customers to integrate remanufactured elevators into their building circularity strategies.

Industry 4.0 - Smart Manufacturing

As manufacturers transition to Industry 4.0 for enhanced product development and quality control, KLEEMANN can stay ahead in a competitive market, focusing on compliance with new regulations and implementing advanced digital manufacturing processes. Simply put, smart manufacturing can be regarded as the collection and connection of data across all production steps in the production line. IoT has already been explored by KLEEMANN in regard to elevator maintenance and the safety of the production line, but compiling and utilizing all available data could be greatly beneficial to the company as a whole if implemented in the production industry. By implementing enhanced digital

technologies, KLEEMANN can advance product quality, improve flexibility in productivity, increase production capacity, manage quality through market expansion, lessen its environmental impact, and manage costs, setting itself apart from its competitors.

The emergence of Industry 4.0 has provided rapid development of digital transformation in manufacturing technologies synonymous with smart manufacturing (Psarommatis et al., 2022). Industry 4.0 involves increasingly using data and simulation capabilities to make informed decisions. The EU-funded project "Data4.Zero" involves 20 European partners collaborating to improve the quality of products from European manufacturers (Powell et al., 2022). The project aims to provide companies with a digital quality assurance system to collect and analyze data from manufacturing processes. This digital capability will enable companies to have better control over their production, with the ability to predict and prevent defects through state-of-the-art digital methods that impact a total product's life.

Other benefits of the addition of Industry 4.0 into aspects of manufacturing include an increase in production capacity and productivity. Global management consultants, such as McKinsey and Company, are proponents of digital transformations in this field and predict that Industry 4.0 improves manufacturing labor productivity by 15-30%, and reduces machine downtime by 30-50% (Gregolinska et al., 2022). Not only do these advances positively impact the environment by decreasing a product’s individual production timeline, but they also boost the efficiency of the entire manufacturing process.

Implementing IoT systems within the manufacturing process allows KLEEMANN to make more informed decisions by leveraging data gathered from each production step in the line. This data can not only be analyzed to give KLEEMANN better control over production

processes, but data-driven solutions also offer an 85% increase in overall forecasting and prediction accuracy (Gregolinska et al., 2022).

Smart manufacturing is a revolutionary approach, however, it comes with challenges. For smart manufacturing to be implemented, sensors need to be put at every step of the production line so data at every step can be gathered. This data then needs to be stored all together in a central database that is easily accessible and can be analyzed. In order for smart manufacturing to be truly effective, the data tracked by the sensors needs to be accurate and current. Errors and unwanted variations in manufacturing must be identified and flagged immediately to enable quick intervention and resolution. Implementing smart manufacturing comes with a steep initial cost, as KLEEMANN must buy sensors and other data-gathering equipment. Despite these challenges, the long-term benefits make it a worthwhile endeavor for forward-thinking manufacturing companies like KLEEMANN.

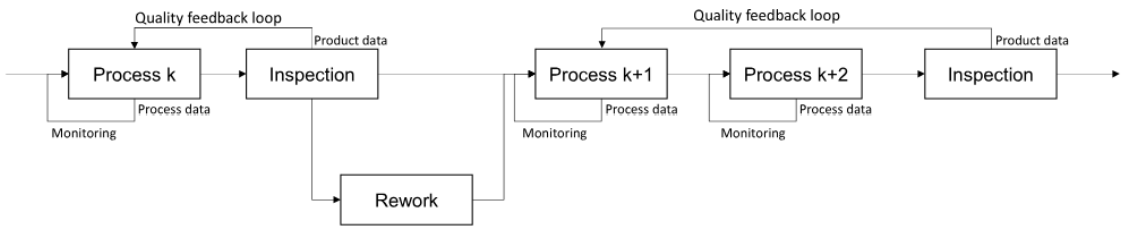
Zero Defect Manufacturing

As an early adopter of digital quality methods from the "Data4.Zero" project, KLEEMANN can utilize data to implement zero defect manufacturing (ZDM) to optimize production, improve the design further, and enhance their LCM processes, leading to faster delivery, competitive prices, and more rapid development and implementation of sustainable solutions. The shift to sustainable production requires manufacturing companies to produce higher quality products with increasing complexity at lower cost while minimizing resource use and waste. Prioritizing quality can lead to reduced costs and better resource utilization. Zero Defect Manufacturing (ZDM) aims to achieve "first-time-right" quality and reshape manufacturing ideology by integrating production planning, quality management, and maintenance management. ZDM is an emerging paradigm beyond traditional quality approaches such as Lean Production and Six Sigma, which is based on a single feedback loop

process strategy, leading to detection and repair, see Figure 10 (Powell et al., 2022). ZDM aims to eliminate defects by detecting and correcting defective products and process parameters and predicting and preventing defects, see Figure 11 (Powell et al., 2022).

Figure 10

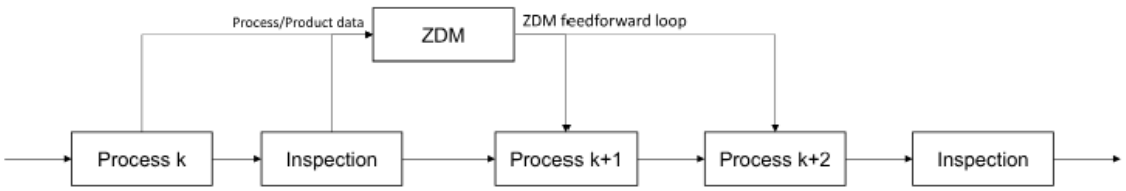
Schematic Representation of Lean Production and Six Sigma Quality Strategies



For instance, the fundamental principle of Six Sigma is that there are 3.4 defects per 1 million products, which is close to zero. ZDM also incorporates similar principles, but the distinction lies in how it is attained through statistical analysis as opposed to prediction using digital technology (Psarommatis et al., 2022).

Figure 11

Schematic Representation of ZDM Strategies



Barriers and Opportunities to the Implementation of ZDM

Barriers that can impede the implementation of ZDM include the expenses and the amount of time that it can take for a thorough analysis of different components, especially in

the beginning of the system implementation. Implementing new systems can be costly, including operational and upkeep costs, training costs, and initial investment costs. After some time of operation, though, the system will begin to help KLEEMANN generate more money than it is costing them to upkeep.

The main opportunities that present themselves with the implementation of ZDM are detection, repair, prediction, and prevention (Psarommatis et al., 2020). Detection can be a resource in ensuring that, before it even reaches the manufacturing point, the elevator components are of the highest quality and meet all of the company's standards. ZDM will also enable KLEEMANN to determine if a product can be repaired when a defect occurs during the current manufacturing stage. Predicting issues before they arise is something that will aid KLEEMANN in the ZDM process, knowing what they need to fix before sending it through a thorough analysis to save time and money in using the system. Lastly, looking into preventative measures and maintenance, which goes hand-in-hand with the predictive process. Using both predictive and preventative measures will decrease the time of analysis for each part of the elevator, as well as creating better quality control from the beginning.

The ZDM implementation process works best when prioritizing the parts of the elevators that KLEEMANN finds to be higher risk, such as the parts that have the highest impact if defected, to ensure that the analysis does not slow production in the company (Psarommatis et al., 2023). This allows for a sort of organization when using ZDM, focusing on items that potentially have a greater impact on the functionality of the elevator. Once implemented, although there is a risk of ZDM being costly and time consuming in the beginning, it is expected to yield a higher payout based on the increased quality.

Way Forward for Implementation of ZDM at KLEEMANN

Implementation of ZDM at KLEEMANN would be beneficial to learn about what components of the elevator are malfunctioning sooner, and how that specific item can be improved to then cause less defects in future elevators. KLEEMANN already utilizes software to detect wear and tear of the elevator during its lifetime, but using ZDM will focus on the manufacturing process of the elevator. Finding the dilemmas of the manufacturing process of an elevator before it is complete will allow for less mistakes which will then lead to less lost product and money. Because there will be a significant decrease in defects of products, this will significantly reduce the carbon footprint and waste found in the environment.

KLEEMANN has already begun to implement resources that will help the ZDM system, such as a machine that cuts specific wires to be a certain length and in a certain quantity. Machines such as this erase some human error, eliminating another aspect that the ZDM system would need to spend time analyzing. Implementing this type of technology in other aspects of the company would help to increase efficiency and combat production error, overall increasing the quality of KLEEMAN products.

Lessons Learned

Global engineering design poses unique challenges and opportunities, requiring a multidimensional approach that considers technical aspects and cultural, social, and environmental factors. Through engaging in a project on global engineering design, valuable lessons were learned about the importance of collaboration, cross-disciplinary communication and cooperation, cultural awareness, sustainability practices, and stakeholder engagement.

One key lesson from this project was the critical role of collaboration in achieving successful outcomes for complex projects. The stakeholders may come from diverse backgrounds with varying perspectives affecting their desired goals and objectives. Therefore, it is essential to incorporate open communication channels among team members through regular meetings while considering differing perspectives and opinions on how to proceed.

Projects Limitations

These projects are limited by the time constraint of 12 days from initiation to delivery. With a condensed schedule, there may be limited opportunities to address all relevant aspects of these projects. The proposed solutions are not detailed; however, they are designed to be a guide on the multitude of variables that should be evaluated for further detailed design.

Additionally, testing materials and a lab to execute these tests are unavailable; therefore, everything concluded in this report has been theoretically based. With limited materials, all data has been gathered through research without physical experimentation. The proposed solutions are meant to aid in possible experimentations to prove whether the theoretical solution is valid.

Another important limitation was the limited time to learn about the KERS, life cycle management, and the elevators. As it was our first time working with such a topic, research

and time to deepen our understanding of the topic was needed. Moreover, the research on KERS available on the internet was in short supply. As it appears evident, it is a very new topic that is still being researched. Consequently, we had to combine the few existing methods of KERS and life cycle management and apply them on the elevators with creative and synthetic imagination.

Our team is composed of different engineering majors and backgrounds. The multidisciplinary approach can have a positive influence on our project through organization and stimulation of various ideas. Organizing and expressing our thoughts and views clearly can consume a lot of time. Despite our diverse backgrounds and knowledge, we were able to come together to present our findings.

A budget estimate was not provided for this project, so an exact number for the labor required and the installation of the implemented elevators may not be fully accurate when it comes to execution.

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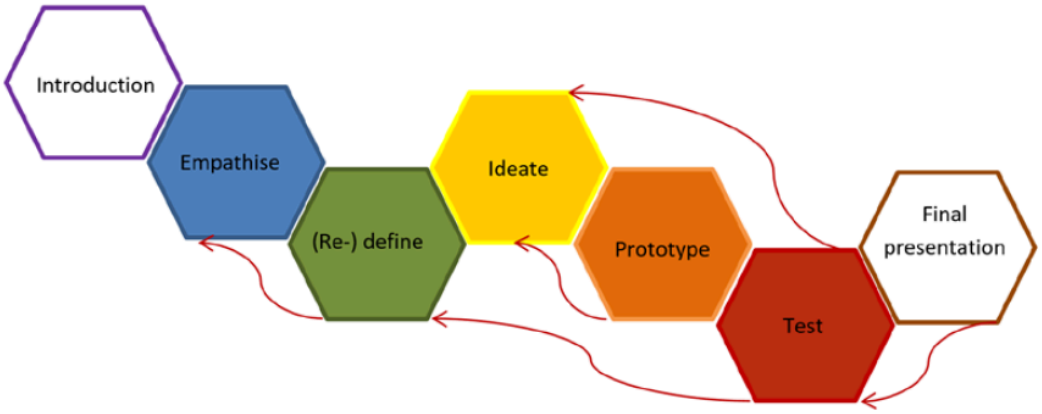
Appendix

Appendix A: Figures

Figure 1

United Nations Sustainable Development Goals (United Nations, 2023).

Figure 2

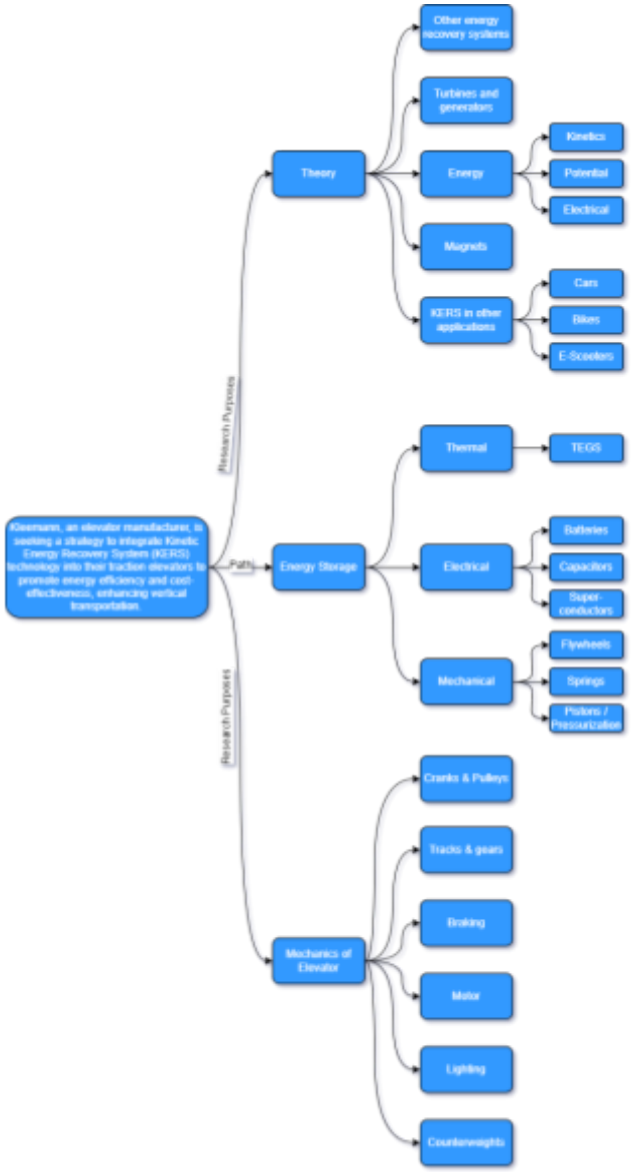


Design Thinking Model

Figure 3

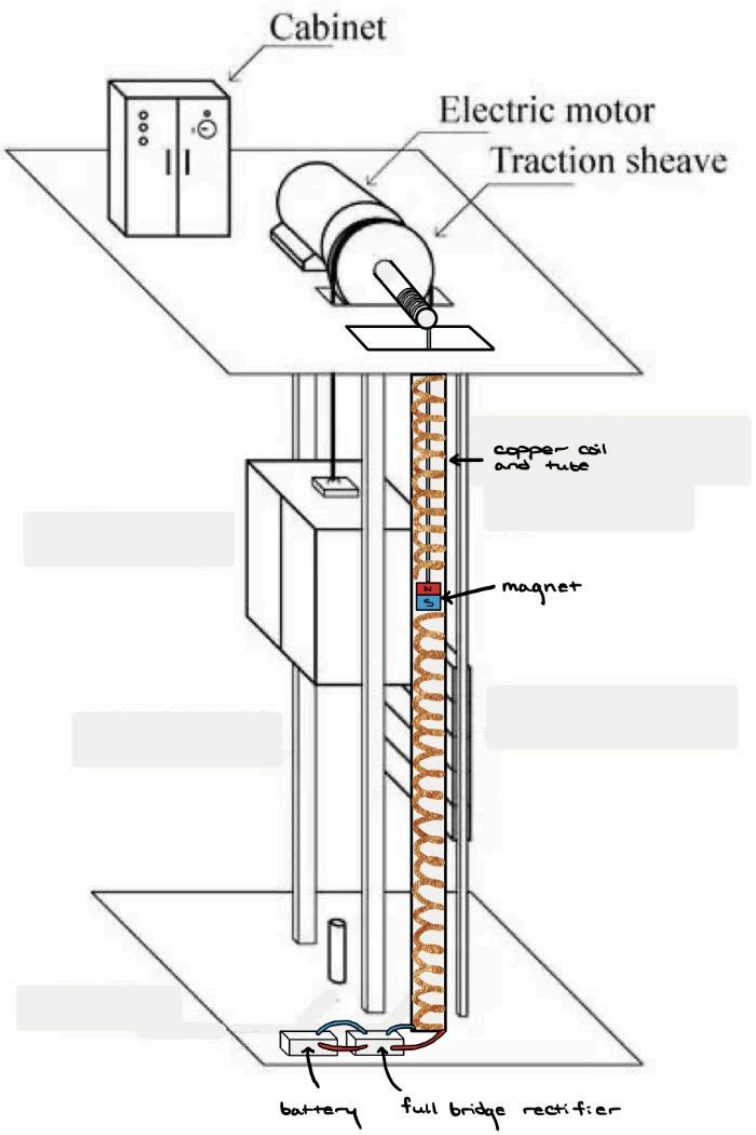
Project Management Life Cycle

Figure 4



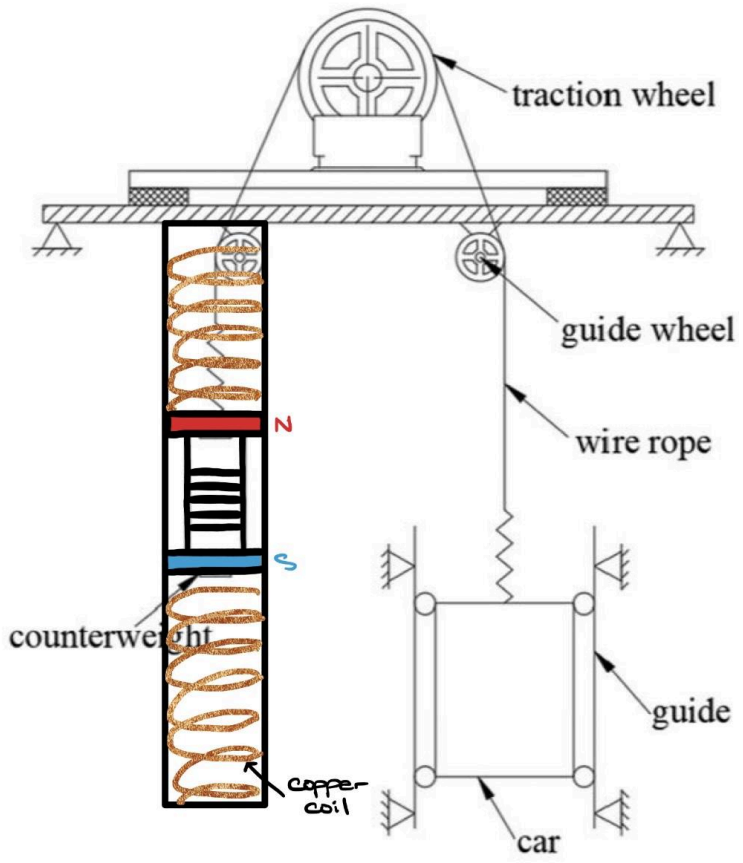
KERS Logic Tree

Figure 5



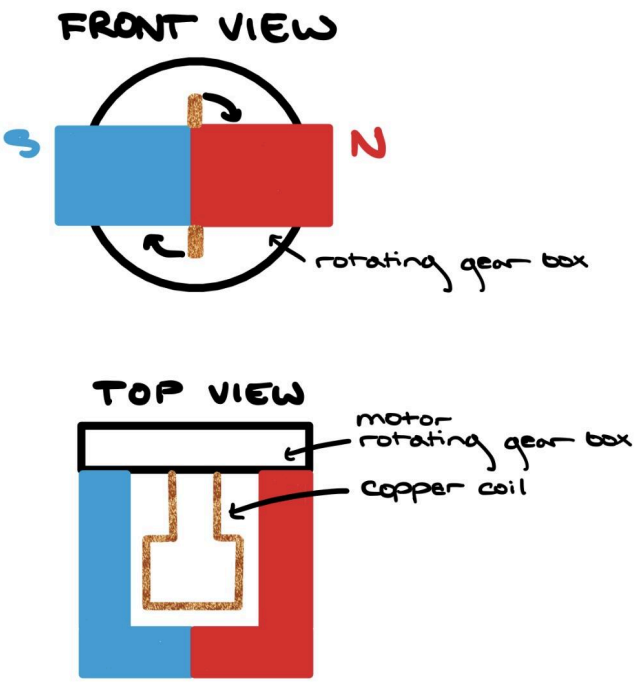
Technical Diagram of KERS Coil/Magnet System (Design 1)

Figure 6



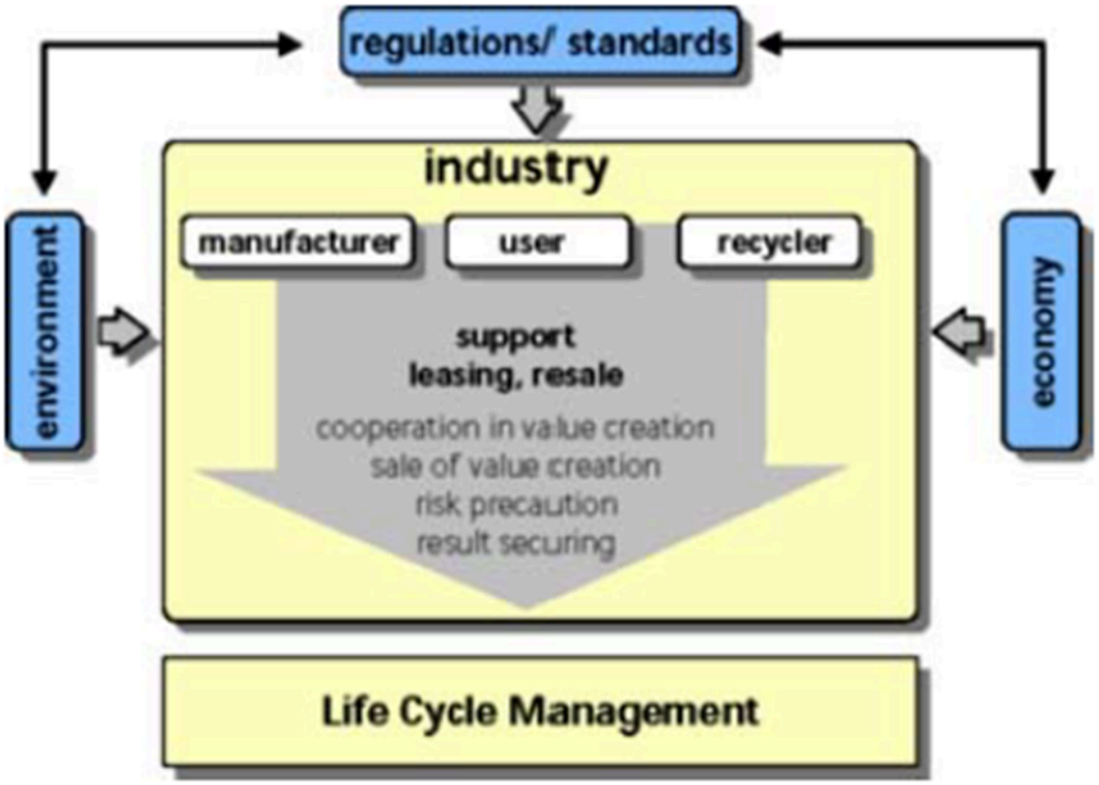
Technical Diagram of KERS Coil/Magnet System (Design 2)

Figure 7



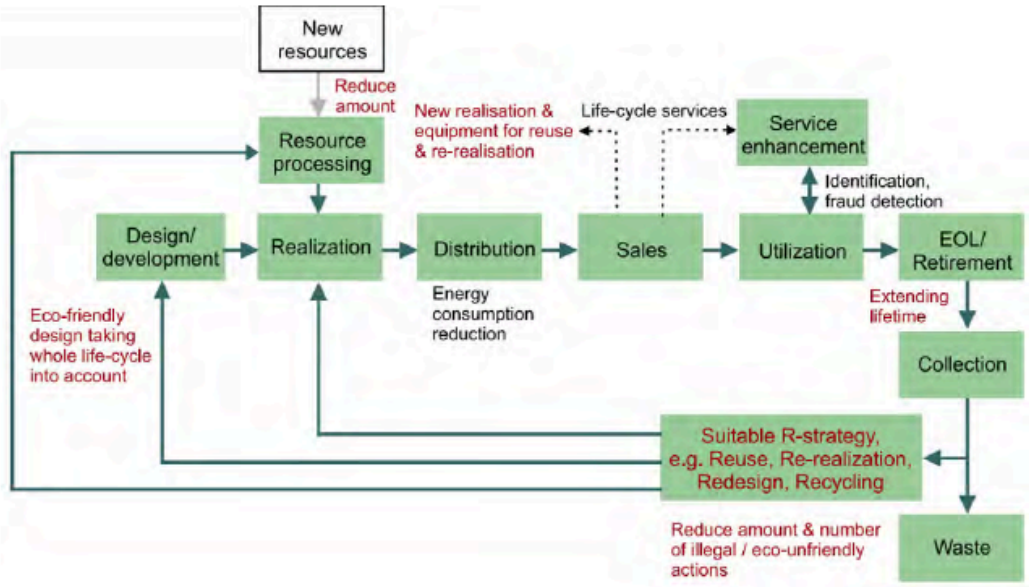
Technical Diagram of KERS Coil/Magnet System (Design 3)

Figure 8



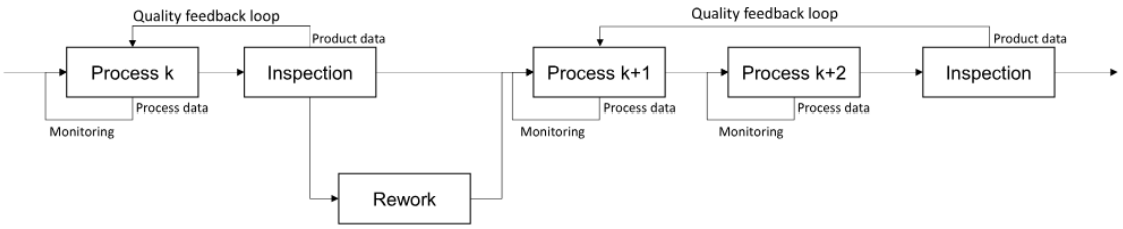
Environment, Regulatory and Economy Demand for Life Cycle Management (Westkämper, 2000).

Figure 9



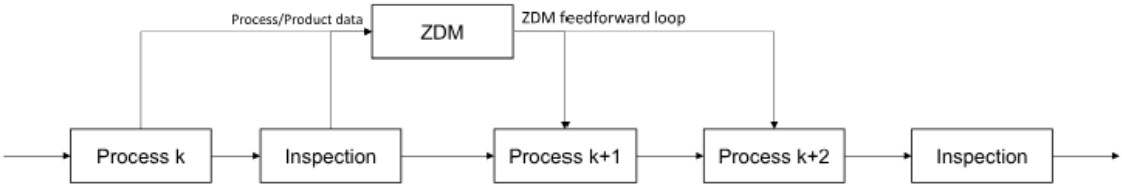
Life cycle model that considers the circular economy (Valkokari et al., 2016).

Figure 10



Schematic Representation of Lean Production and Six Sigma Quality Strategies

Figure 11



Schematic Representation of ZDM Strategies

Appendix B: Tables

Table 1

Comparison of different implementations of coil-magnet energy production system

Potential Implementation	Advantages	Disadvantages
1	• Cost effective (compared to #2) • Separate system makes it easier to add on to existing elevator systems • Easier to maintain	• Requires a lot of wire (compared to #3) • Need additional space • Extra power may be required to move weight of magnet
2	• Magnet weight can be integrated into counterweight (would not need additional power)	• Requires a lot of wire (compared to #1 and #3) • Since it is integrated into elevator, would not be implemented as an “add-on”
3	• Less wire required (compared to #1 and #2) • Cost effective (compared to #2) • No need to create separate pulley system	• Must create spinning shaft off of motor (would not be implemented as an “add-on”) • Extra power required to spin coil

Table 2

Key Parameters for the MACS Analysis

Inputs	**Outputs**
Magnet	Power (W)
Material	Voltage (V)
Magnetic Field Strength (T)	Current (A)
Magnet Radius (m)	Resistance (Ω)
Cost ($)	Wire Length (m)
Coil	Cost ($)

Material
Wire Thickness (m)
Coil Radius (m)
Number of Turns Per Meter
Cost Per Meter ($)
Resistivity ($\Omega \cdot m$)

Appendix C: Code

```
'''
File: MACS_Sample_KLEEMANN.py
Author: Rahul Ayanampudi
Author Title: Texas A&M University Aerospace Engineering Student, ENGR
410 Global Engineering Design, Summer 2024
Sponsor: KLEEMANN
Date Created: 05/28/2024
Description: KERS Team -- Magnet and Coil System (MACS) Simulator
'''

import numpy as np

class magnet:
    def __init__(self, diam, cost, B, material):
        self.radius = inToM(diam / 2) # enter as inches, stored in
meters
        self.cost = cost # dollars
        self.B = B # Tesla
        self.material = material

class coil:
    def __init__(self, wireThickIn, loopRadius, numTurnsPerM,
costPerFt, rho, material, coilLength):
        self.wireThickness = inToM(wireThickIn) # enter as inches,
stored in meters
        self.loopRadius = loopRadius # meters
        self.numTurnsPerM = numTurnsPerM
        self.coilLength = coilLength
        self.cost = getCost(getCostPerM(costPerFt),
getLength(loopRadius, numTurnsPerM, coilLength)) # dollars
        self.R = getResistance(rho, loopRadius, numTurnsPerM,
coilLength, wireThickness) # ohm meters
        self.material = material

def inToM(val):
    return val * 2.54 / 100

def getLength(loopRadius, numTurnsPerM, coilLength):
    return 2 * np.pi * loopRadius * numTurnsPerM * coilLength

def getCostPerM(costPerFt):
```

```
    return costPerFt * 3.048

def getCost(costPerM, l):
    return costPerM * l

def getArea(wireThickness):
    return np.pi * (wireThickness / 2) ** 2

def getResistance(rho, loopRadius, numTurnsPerM, coilLength,
wireThickness):
    return rho * getLength(loopRadius, numTurnsPerM, coilLength) /
getArea(wireThickness)

def getMagTime(length, speed=1): # m, m/s
    return length / speed

def getSystemProps(mag, c, coilLength, time):
    V = ((c.numTurnsPerM * coilLength) * mag.B * np.pi * c.loopRadius
** 2) / time
    I = V / c.R
    P = I * V

    # system outputs
    print('Magnet: ' + mag.material + " " + str(mag.radius))
    print('Coil: ' + c.material + " " + str(c.wireThickness) + " with "
+ str(c.numTurnsPerM))
    print('Power:', P, 'W')
    print('Voltage:', V, 'V')
    print('Current:', I, 'A')
    print('Resistance:', c.R, 'ohms')
    print('Wire Length:', getLength(c.loopRadius, c.numTurnsPerM,
c.coilLength), 'm')
    print('Magnet Cost: $' + str(round(mag.cost, 2)))
    print('Wire Cost: $' + str(round(c.cost, 2)))
    print('Total Cost: $' + str(round(mag.cost + c.cost, 2)))
    print()


# Specification provided by Kleeman
speed = 1 # m/s
```

```
coilLength = 3 # m, adjust this parameter as necessary to constrain the
current to an acceptable level
time = getMagTime(coilLength, speed) # s


# Create magnets
magnets = []
Nd3 = magnet(3, 188.63, 1.48, 'Neodymium')  # cost from McMaster-Carr
assuming bulk order of 50+
Cr3 = magnet(3, 26.60, 0.38, 'Ceramic')  # cost from McMaster-Carr
assuming bulk order of 50+
Nd6 = magnet(6, 229, 1.32, 'Neodymium')  # cost from McMaster-Carr
assuming bulk order of 50+

magnets.append(Nd3)

# create coils
coils = []

Cu12Coils = []
numTurnsPerMList = [750, 1000]
wireThicknessList = [0.1, 0.13, 0.18] # in
wireCostPerFt = [0.18322, 0.43786, 0.90908] # McMaster-Carr for solid
copper wire with 600V AC

for idx1, mag in enumerate(magnets):
    for idx2, wireThickness in enumerate(wireThicknessList):
        for idx3, numTurnsPerM in enumerate(numTurnsPerMList):
            Cu12Coils.append(coil(wireThickness, mag.radius,
numTurnsPerM, wireCostPerFt[idx2], 1.72e-8, 'Copper', coilLength))

coils += Cu12Coils

for mag in magnets:
    for c in coils:
        getSystemProps(mag, c, coilLength, time)
```